

KASANDY FOUNDATION

Kasandy Circular Economy Initiative

MATERIAL RESEARCH PROTOCOL

Acoustic Brick Material Testing: Research Protocol & Objectives

Research framework and testing agenda for textile-based acoustic brick development in partnership with Emily Carr University of Art + Design (ECUAD)

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kasandyfoundation.org

This document is a research protocol. Material testing begins July 15, 2026 following shredder installation. No test results are available at time of publication.

Acoustic Brick Material Testing: Research Protocol & Objectives

This document establishes the research protocol and objectives for KCEI’s acoustic brick material testing program, to be conducted in partnership with Emily Carr University of Art + Design (ECUAD) beginning July 15, 2026. The protocol is issued in advance of testing to establish methodological rigour, align expectations between KCEI and ECUAD research partners, and provide transparency to funders and stakeholders about the research agenda.

KCEI’s industrial textile shredder is scheduled for installation and commissioning at the ECUAD facility by end of June 2026. Material testing will commence on or after July 15, 2026, coinciding with the formal start of the KCEI–ECUAD research collaboration under the Memorandum of Understanding (MOU).

Timeline milestones

- End of June 2026: Industrial textile shredder installation and commissioning at ECUAD facility
- July 15–August 1, 2026: Formal commencement of KCEI–ECUAD research collaboration; initial specimen preparation and baseline testing
- August–October 2026: Primary testing phase — compression, acoustic, dimensional, and humidity cycling
- Q4 2026: Interim findings report circulated to ECUAD research leads and KCEI team
- 2027: Full results report; commercial specification development pending satisfactory outcomes
- 2027–2028: Extended durability and fire resistance testing as required for certification pathway

SECTION 1: RESEARCH CONTEXT

Research Context & Rationale

The global precedent for textile-based acoustic construction materials was established by FabBrick (Paris, est. 2018), which has demonstrated that shredded textile fibres, compressed with an ecological binder, produce non-structural bricks with both acoustic and aesthetic properties. FabBrick’s artisanal production has processed approximately 12 metric tonnes of textile waste since 2018 — an average of 2 tonnes per year. KCEI’s industrial approach is designed to achieve this volume in weeks rather than years.

The KCEI–ECUAD research collaboration will generate the first formally peer-reviewed Canadian evidence base for industrial textile-to-brick conversion. By embedding material science rigour from the outset, KCEI aims to establish a product specification that satisfies commercial procurement requirements and supports a longer-term certification pathway.

Why this research matters

- Establishes KCEI product performance data that funders, municipal procurement officers, and commercial buyers can rely on
- Builds the evidence base for certification applications (required for construction industry commercial deployment)
- Positions KCEI as a research-backed initiative, not merely a demonstration project
- Generates publishable findings that strengthen KCEI's grant applications and municipal partnership proposals

SECTION 2: RESEARCH OBJECTIVES

Research Objectives

The research program addresses four primary questions about KCEI acoustic brick performance:

Objective 1: Compression performance

What is the compressive strength of KCEI acoustic bricks under varying fibre composition and binder ratio conditions? What failure mode do specimens exhibit? Is performance sufficient for non-structural interior partition and acoustic panel applications? This data will form the basis of the product's structural specification.

Objective 2: Acoustic performance

What sound absorption coefficient (NRC) do KCEI acoustic bricks achieve across the frequency range relevant to interior acoustic applications (250 Hz to 4000 Hz)? How does acoustic performance vary with specimen thickness, fibre composition, and compression density? How does performance compare with commercially available foam and fibre acoustic panels?

Objective 3: Dimensional stability

Do KCEI acoustic bricks maintain dimensional consistency across the humidity and temperature variation typical of BC interior environments? What level of dimensional variation is expected across standard production batches? Is variability within acceptable tolerances for commercial installation?

Objective 4: Visual and surface quality

What is the range of colour variation achievable through textile input sorting? Is surface texture and appearance consistent enough for commercial interior applications? What quality control protocols are required at the production stage to achieve specification-grade output?

SECTION 3: METHODOLOGY

Proposed Methodology

Specimen preparation protocol

Post-consumer textile waste will be shredded using KCEI's industrial shredder to produce fibres of 10–40mm length. The shredded fibre will be blended with an ecological binder at three ratios (10%, 15%, 20% binder by weight) and compressed in standard moulds. Two standard formats will be produced: standard brick (200mm × 100mm × 50mm) and panel (400mm × 200mm × 30mm).

A minimum of 5 specimens per binder ratio per format will be produced for each testing protocol, providing sufficient replicates for statistical analysis.

Testing protocols

- Compression testing: Axial load testing using calibrated hydraulic press at ECUAD facility. Controlled loading rate of 2mm/minute. Specimen failure or 10kN load limit.
- Acoustic measurement: Normal incidence sound absorption coefficient using impedance tube (Kundt's tube) method, 250–4000 Hz range. Random incidence measurement (reverberation room method) planned for Year 2.
- Humidity cycling: 14-day cycling at 30–80% RH, dimensional measurement before and after
- Visual assessment: Structured scoring rubric for colour consistency, surface texture, and edge quality

SECTION 4: ECUAD PARTNERSHIP ROLES

ECUAD Partnership Structure

The research is conducted under the formal KCEI–ECUAD Memorandum of Understanding. Key roles:

- Ellen [Surname], Faculty Lead: Research design, academic oversight, grant applicant for ECUAD contribution
- William [Surname], Facilities: Equipment access, laboratory space, installation coordination for shredder
- Annika [Surname], Postgraduate Researcher: Day-to-day specimen preparation, testing execution, data collection and analysis
- KCEI (Deress Asghedom, Ops Lead): Textile input supply, shredder operation, manufacturing protocol development
- KCEI (Jackee Kasandy, Executive Director): Strategic oversight, funder reporting, commercial application development

Expected Outcomes (2026–2028)

This research program is expected to produce, over its 2026–2028 horizon, the following outputs:

- Year 1 (2026): Baseline performance data for acoustic brick compression, acoustic absorption, and dimensional stability; interim findings report
- Year 2 (2027): Full results report; commercial product specification sheet; initial client conversations with institutional buyers

- Year 3 (2028): Assessment of certification pathway viability; fire resistance testing if earlier results are satisfactory; peer-reviewed publication of findings

The research is designed to produce commercially actionable results, not merely academic outputs. Every testing decision will be evaluated against the question: what does a municipal procurement officer or commercial buyer need to know to specify this product?

KCEI–ECUAD Research Partnership. Shredder installation: end of June 2026. Research commencement: July 15, 2026. Contact KCEI: hello@kasandyfoundation.org. Contact ECUAD: through Ellen [Surname], Faculty of Design. kasandyfoundation.org.